

system's screen. Theoretically, there is no limit to the number of clients that can be connected. Isn't it an awesome concept?

Synergy supports all major platforms and works fine with MS Windows, GNU/Linux and Mac OS X, allowing the user to seamlessly move the mouse cursor between multiple sessions. The operating system requirements for Synergy are given below:

- MS Windows: Windows XP3 onwards
- Ubuntu: Ubuntu 10.04 onwards
- Mac OS X: Mac OS 10.5 onwards

But if we can also share the keyboard and mouse by using a KVM switch, then what is special about Synergy? Well, KVM is hardware; hence, you have to purchase it, which adds to costs and also occupies some space on your desk. On the other hand, Synergy is purely a software component. Hence, it doesn't occupy space on your desk. Some of the salient features of Synergy are listed below:

- 1) It is a lightweight application with a very low footprint.
- 2) It is open source and entirely free of cost. As its source code is available to the public, you can customise it according to your requirements.
- 3) It is cross-platform and works fine with all major OS platforms.
- 4) It has full encryption support, which keeps sensitive data safe on public networks.
- 5) One of the great features of Synergy is that it shares the clipboard between Synergy sessions.
- 6) It allows one to move the mouse seamlessly between Synergy sessions.
- 7) It doesn't require any special hardware. Only network connectivity is required.
- 8) It doesn't require a high-speed network and works seamlessly with regular network bandwidth.

Installation steps

Now that we are familiar with the use cases of Synergy, let us quickly go through the installation steps. The section below describes the installation procedure on Ubuntu GNU/Linux and MS Windows. For any other option, refer to the installation guide of the OS you're interested in.

Synergy is available in Ubuntu's official software repository. You can install it by using the following command:

```
[bash]$ sudo apt-get update
```

```
[bash]$ sudo apt-get install synergy
```

In addition, you can install a pre-compiled .deb package. To download the latest version of Synergy, visit its official website <http://synergy-project.org/> and download the appropriate version. The steps below describe the installation of the .deb package on Ubuntu using the *dpkg* package manager.

```
[bash]$ ls
synergy-1.5.1-r2398-Linux-x86_64.deb
```



Figure 1: Synergy diagram

```
[bash]$ sudo dpkg -i synergy-1.5.1-r2398-Linux-x86_64.deb
```

For MS Windows, download the pre-compiled installer from the same location. Double click on the installer and follow the on-screen installation instructions.



Note: Please ensure that versions of both the Synergy applications are the same.

Configuration steps

It takes only a few minutes to configure Synergy. Configuration is required only at the Synergy server (hereafter referred to as just the server) side, and Synergy clients (hereafter called clients) can directly connect to it.

First, choose a server. Obviously, we will choose Ubuntu as the server because the external keyboard and mouse are connected to it. We can configure the server either by creating a configuration file or by using a GUI tool. This section describes both methods.

The server machine shares its keyboard and mouse with clients. It needs to know about all the clients that will connect to it. So, to inform the server about these clients, create a configuration file. Synergy's configuration file is a plain text file with two mandatory sections and one optional section. The configuration file for our set-up will look like this:

```
[bash]$ cat ~/synergy.conf
section: screens
ubuntu:
windowspc:
end

section: aliases
windowspc:
xxx.xxx.xxx.xxx # Replace it with actual IP of Windows machine
end

section: links
ubuntu:
left = windowspc
windowspc:
right = ubuntu
end
```